

Digital Archives & Methods Portfolio

Presented now are the assignments done throughout the semester.

RegEx & OpenRefine

Task 1: List regex that extracts dates from <http://bit.ly/regexexercise2> and formats them as YYYY-MM-DD?

URL for the solution: <https://regex101.com/r/A0UC0D/2>

Brief explanation:

The regex used is: `(\d{1,2}).(\d{1,2}). ?(\d{4})`

Through the (parenthesis) we've grouped each part of the date which is in the original text in the format: Month, Day, Year. The `\d` catches digits while the {curly brackets} specify the amount of numbers in it. The `.` and `.?` are there to catch numbers spaced oddly or split by comma.

The substitution used is: `{ $3 } { $1 } { $2 }`

Here we simply reorganize the groupings we've already defined into the proper year, month, day format as asked. Thus we have our solution.

Task 2 Regex to convert stopwordlist between R and Voyant format

R to Voyant: Our Regex is: `^c(|\l)$|",\s*` and our Substitution is: `\n`

Brief explanation:

The regex subsequently catches the commas, quotations and spaces while leaving the words untouched. Then substituting with `\n` we create a newline and create a stoplist suited for Voyant.

Voyant to R: Our Regex is `\n` and our substitution is `“ , “`

Brief explanation:

The regex being a plain `\n` catches each new line while leaving the words untouched. Our substitution simply wraps each word in a quotation and separates them by commas. Using this method one will have to manually add a quotation at the start of the very first word and at the end of the very last word. However this is a small task that does not take much time.

Task 3 Does OpenRefine alter the raw data during sorting and filtering?

It does not.

Task 4: What are the two most water-deprived months in Mozambique and Tanzania?

We edit the data. Transforming the cells with: `value.replace("[", "").replace("]", "").replace("'", "").replace(";", "")` to unify the cells and remove unwanted quotations and brackets. Giving us a cleaner list. We also do: `value.replace(" ", "")` to remove spaces since otherwise they'll count "Oct" and " Oct" as two separate entities. Having this final list we can conclude that the two most water-deprived months are: October (74) and September (70).

Task 5: Challenge - 10 most common occupations among in 1801 Aarhus

- Total number of people in your selected cohort: 3311, I have focused on men.
- Define what is / is not an occupation:

I have gone from the principle that an occupation is their primary source of income. Given limited time the data presented below may be wrong and/or different from others, however that is the joy of working with datasets like these and being able to 'fix' the data. Given much more time I would be able to properly sort and unify the various jobs into one and get much more sufficient data. Suffice it to say OpenRefine is a powerful tool but I am still learning it.

The following list is based on men in the age of 20-30

- 10 most frequent occupations (list number of people practising that job with each entry)
1. Soldat (389)
 2. Bonde og gaardbeboer (117)

3. Tjenestekarl (37)
4. Væver (27)
5. Huusmand med jord (22)
6. Gårdskarl (11)
7. Gaardmand (10)
8. Land recrut (10)
9. Læredreng (10)
10. Matros (10)

Tidy Spreadsheets and the Danish Succession

This week we began working on spreadsheets.

The goal was to create a tidy spreadsheet listing the names of Danish monarchs. This had to include their birth- and death-year and the start and end of their reign.

Given scheduling troubles, Manuel and I decided to make each our own spreadsheet. As such the following spreadsheet and statements should be attributed to Mathias Salling Pedersen.

Under the submitted assignment you will find the file: 'Danish Monarchy'. This is the spreadsheet created. In it you will find each Danish monarch, their birth- and death-years as well as the years they reigned. The odd monarch has a note attached to them that explains any oddity regarding their reign. (For example the King who ruled twice.)

This spreadsheet was made entirely by hand whilst crossexamining the sources we'd been given. This being the website we were recommended. It can be found [here](#).

The spreadsheet itself has been uploaded alongside this document.

What are the basic principles for using spreadsheets for good data organisation?

There are a few guiding principles when creating a spreadsheet that one can adhere to in order to create a tidy and organised dataset. Firstly let's discuss **Multivalued Attributing**. Excel cells typically only handle one value. (EG: "Blue" and "Red" rather than "Blue, Red") To create a tidy dataset it is recommended to keep cell values at one. Additionally **Consistency is Key**, make sure to have consistent cell values. (EG: "No." should be "No" the entire way through. Not "no" or "N" or other such examples.) In the "messy" example we were shown we also took note of multiple tables per worksheet, this might confuse programs like **R** and thus one worksheet should hold one table. If we come across missing data do not use "0" but rather "NA" without the slash. As was said

in class: **ambiguity is our biggest enemy**. Furthermore consistency and completeness of data can be improved through **Data Validation**. The function found in Excel allows you to set certain parameters, such as defining what data is 'allowed' in a section. Thereby we can create a list of accepted terms and the spreadsheet will inherently reject any cells filled with terms that are not on said list. When you have a well made spreadsheet you can export the data to formats like **CSV** or **TSV**, allowing it to be used consistently by most programs. Thus we can also take **CSV** files and convert them into readable spreadsheets. It is important to note that being Europeans with predominantly European operating systems we use the **CSV2** format.

Sentiment Analysis: Game of Thrones

Presented is the coding done in Rstudio and the results of them:

```
got_path <- here("data", "got.pdf")
got_text <- pdf_text(got_path)
```

Hide

```
got_df <- data.frame(got_text) %>%
  mutate(text_full = str_split(got_text, pattern = '\\n')) %>%
  unnest(text_full) %>%
  mutate(text_full = str_trim(text_full))
```

Hide

```
got_tokens <- got_df %>%
  unnest_tokens(word, text_full)
```

Hide

```
got_wc <- got_tokens %>%
  count(word) %>%
  arrange(-n)
got_wc
```

This provided us with the most common words used:

word	n
<chr>	<int>
the	17988
and	8997
to	6640
a	6461
of	6200
he	5166
his	5161
was	3904
her	3659
you	3267

1-10 of 11,826 rows

Previous 1 2 3 4 5 6 ... 100 Next

As is oft the case, the most common words tell us little. Thus we used a stoplist to get the proper results we can use for analysis:

```
got_stop <- got_tokens %>%  
  anti_join(stop_words) %>%  
  select(-got_text)
```

Hide

```
got_swc <- got_stop %>%  
  count(word) %>%  
  arrange(-n)
```

Hide

```
got_no_numeric <- got_stop %>%  
  filter(is.na(as.numeric(word)))
```

Hide

```
length(unique(got_no_numeric$word))
```

```
[1] 11209
```

Hide

```
got_top100 <- got_no_numeric %>%  
  count(word) %>%  
  arrange(-n) %>%  
  head(100)  
got_top100
```

Providing us with this more useable list of the most common words:

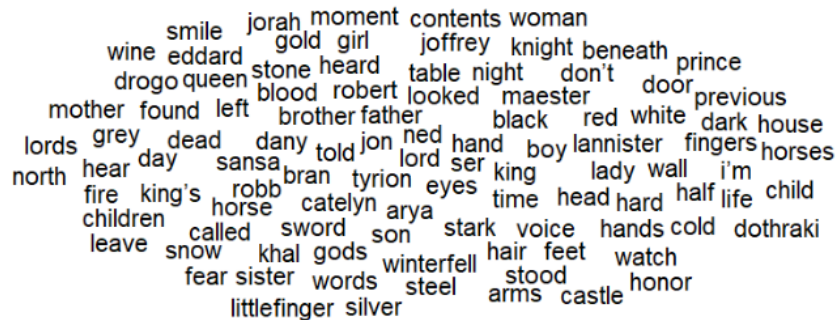
word	n
<chr>	<int>
lord	1341
ser	1023
jon	787
ned	743
tyrion	591
eyes	567
hand	567
king	542
father	512
told	504

1-10 of 100 rows

Previous 1 2 3 4 5 6 ... 10 Next

Next we create a word bubble using the following code:

```
got_cloud <- ggplot(data = got_top100, aes(label = word)) +  
  geom_text_wordcloud() +  
  theme_minimal()  
  
got_cloud
```



Let's make it look nicer.

```
ggplot(data = got_top100, aes(label = word, size = n)) +  
  geom_text_wordcloud_area(aes(color = n), shape = "diamond") +  
  scale_size_area(max_size = 20) +  
  scale_color_gradientn(colors = c("darkgreen", "blue", "red")) +  
  theme_minimal()
```



Next we use the afinn lexicon to acquire the sentiments inherent to each common word:

```
get_sentiments(lexicon = "afinn")
```

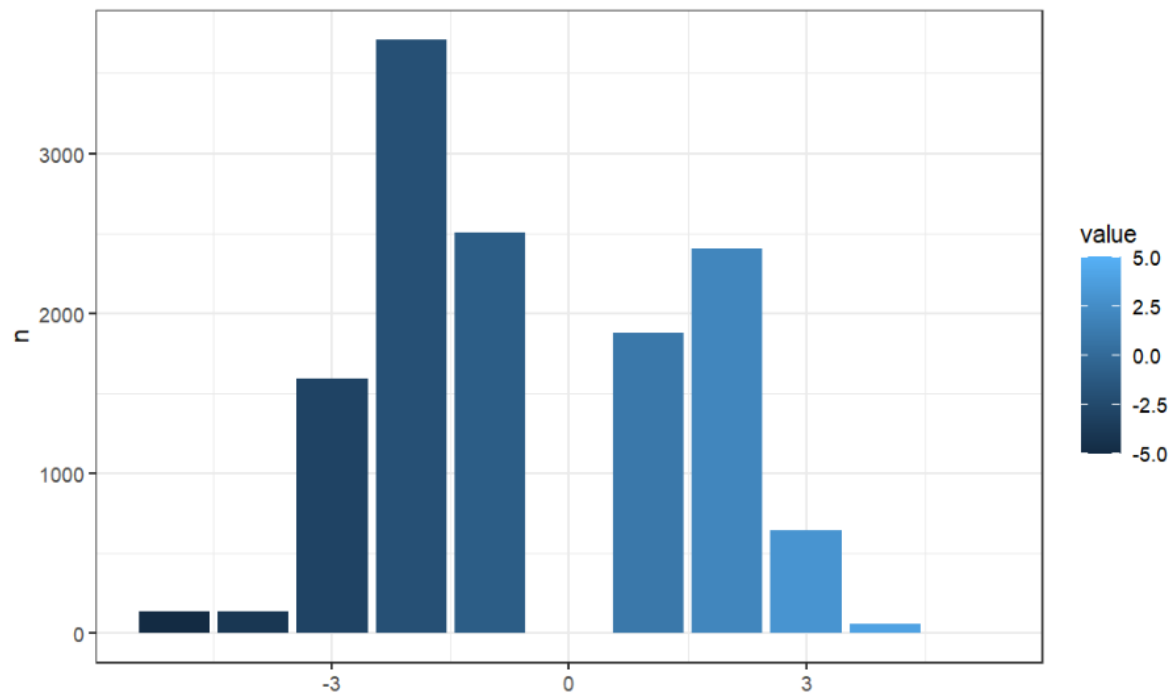
word	value
<chr>	<dbl>
abandon	-2
abandoned	-2
abandons	-2
abducted	-2
abduction	-2
abductions	-2
abhor	-3
abhorred	-3
abhorrent	-3
abhors	-3

1-10 of 2,477 rows

Previous 1 2 3 4 5 6 ... 100 Next

As each word has a positive or negative value, we can analyze the general sentiment of positivity or negativity in the text of A Game of Thrones.

```
got_afinn_hist <- got_afinn %>%  
  count(value)  
  
# Plot them:  
ggplot(data = got_afinn_hist, aes(x = value, y = n)) +  
  geom_col(aes(fill = value)) +  
  theme_bw()
```



Unsurprisingly, the sentiment of A Game Of Thrones is primarily negatively connotated. This makes sense as the fictional story is one of strife. It presents a bleak fantasy world

steeped in realism and inspiration from real world history. The real world War of the Roses being a primary inspiration for the fictional War of the Five Kings.

Digital Archives Reflections Assignment

Part 1

Explain how Jensen's (2021) category of "digital archival literacy" addresses the distinct challenges digital archives represent for historical research. How has digitization changed historians' interactions with their sources? In your response, focus on two points that you found most thought-provoking. You may also choose to cite Jensen and Schriver (2022) in your response. (250 words)

The introduction of digital archives and the lack of literacy that many historians have regarding it introduce several challenges. In Jensen's article we are notified that recent work by scholars in Information Science confirms that a majority of historians use these digital research tools, herein mentioned finding aids, search engines, catalogues, etc. However, they fail to consider their impact on methodology. Moreover, as the digitization of sources is funded by investors it tends to favor what is more "popular" or rather in high demand. This can sideline the interests of researchers as we so oft seek to branch out and create new history rather than reiterate on what has already been done. As external funders only want to pay for what is already popular this does introduce a bias in the (sub)-fields of research. However, we now live in a world where using these digital methods are nigh required and it is not necessarily a bad thing. A well-designed archive can mean more accessible resources for all interested. However, we must mention how we used them and why we use them. Failing to do so threatens to make the value and impact of these digital tools invisible and obscured. As knowledge of archives is essential to the work of all historians, no matter the subdiscipline, it is paramount that we understand the internal logic of how an archive is made. This has changed with digital archives and, in some cases, become easier. However, understanding the

differences in how an analog archive sorts their sources versus a digital archive will be a much needed skill for those historians wishing to thrive in their chosen fields in this new digitized landscape.¹

Part 2

Navigate to the following page entitled “David Boder: from wire recordings to website.” <https://ranke2.uni.lu/u/boder/#c-clip.en>. Familiarize yourself with the page and watch the animation “David Boder's interview collection: from analogue to digital” to learn about Boder’s work and recent efforts to digitize his interviews.

- i. Navigate to <https://iit.aviaryplatform.com/collections/231> and select one of David Boder’s interviews to analyze. Briefly provide the biographical details of the interviewee. After reading the “description” panel, discuss the provenance of the source. For example, what metadata is provided by the archive, and which historical details did the archive choose to emphasize in their written summary of the recording? (250 words)

The interview chosen is “David P. Boder interviews Adolph Heisler, August 27, 1946, Geneva, Switzerland.” In the description given there is a primary focus on Adolph Heisler, the interviewee, and his own experiences. Focus on Heisler’s presumption that the Hungarian officials were most likely major participants in the procedure of ghettoizing the subset of Slovakian Jews that did not get transferred with the majority, which Adolph was a part of. Later he was sent to work in the Jawiszowice coal mine which supplied the factories in Auschwitz. He recalls this being less unbearable than the camps in Germany he was later sent to, despite the hellish conditions. There is a focus on lived experience and the horrors felt firsthand by the interviewee. The website itself provides a transcript with an implemented autoscroll which allows the listener to follow along with the interview even if they do not know the language. It seems from this and the preliminary introductory animation that David P. Boder had a focus on the lived experience of the previously incarcerated he

¹ Helle Strandgaard Jensen (2020): Digital Archival Literacy for (all) Historians, Media History, DOI: 10.1080/13688804.2020.17790

interviewed. Given the time frame especially, this interview in particular only being a year after the end of the war, it makes a lot of sense. The pain was lived but still fresh to those who had lived it. To the more privileged it was as of yet unknown what depravities the prisoners had suffered. It is a narrative driven by the individual. David Boder inquires about specific details of the travails and troubles of the individual, providing a focus on their story over generalizing the pain.

- ii. Listen to a few minutes of your chosen interview. Focus on the intonations and emotions expressed by Boder and the interviewee (it's okay if you do not understand German or Russian). Based on a close listening of the speakers, contemplate how such digitized interviews might provide information absent from text-based interview transcriptions. What are the implications of using interview recordings as a historical source? You may choose to draw on Presner (2024) or information from the Ranke.2 website in your response. (250 words)

The tone of the interview itself is calm. It is easy to assume that someone who has undergone the horrors of the Nazi regime firsthand would be emotionally charged, yet Adolph Heiser carries himself calmly and explains his story in detail. David Boder carries the interview with a professional tone and asks many clarifying questions that seek to paint a fuller picture. These interviews can provide more than text-based transcriptions alone. Through being able to hear these wire recordings we get a broader sense of the interviewee themselves. Through Adolph Heiser's unshakeable calm we understand more of him as a person. Though hearing these accounts firsthand we are more likely to carry empathy for their pain suffered.

At the same time, the emotional weight of these recordings can also influence the listener's interpretation of the source. Hearing the voice of a survivor creates a far more personal experience than simply reading a written account. In the pauses and hesitation and the tone chosen he communicates emotions and experiences that are difficult to capture through transcript alone. Because of this the recordings function not only as historical documentation but also as individual human testimony.

However using these interviews as a historical source must be done with critical thinking. The stories of a lone individual seldom speak for an entire people. It must be understood inherently that the interviewee's story is theirs alone and can help broaden an image that must be grounded in macro historical facts. Memory itself can also be imperfect especially when recounting trauma months and years after they occurred. Personal testimony works best when combined with broader historical research.

Part 3

Compare and contrast the use of digitized historical sources in web-based databases with reborn digital sources in the archived Web. Should these sources be treated differently when conducting historical research? To complete this task, draw on

your knowledge of at least two database collections discussed in class and/or your readings. In your response, please support your position using relevant literature from our course, such as Kelley (2019), Johnson (2018), Brügger (2019), and the articles mentioned in part one of this assignment. (500 words)

The historian Niels Brügger argues that using digital sources, whether reborn or born digitally requires a change of methodology. The argument he makes is that a critical viewpoint is required to handle these sources properly on the basis of how digital archives are constructed and curated through technology. Unlike traditional archives, digital archives are not neutral containers of information but systems shaped by preservation choice, metadata structure and software.²

Digitized historical sources and reborn digital sources share similarities in that both have gone through processes of selection and categorization. However, they also differ significantly in how historians must approach them critically. Databases such as SlaveVoyages.org or the USC Shoah Foundation archive largely consist of historical material that originally existed in analog form before being digitized. These databases provide historians with unprecedented accessibility and the ability to analyze large quantities of data efficiently when paired with programs such as RStudio. In the case of SlaveVoyages, historians can perform quantitative analysis of patterns present in the transatlantic slave trade across centuries in a fraction of the time it would have taken previously.

At the same time, digitization can create an illusion of objectivity. Johnson argues that digital representations of enslaved people risk reducing the dehumanizing logic of slavery itself by reducing human life to numbers and cargo. Structured databases simplify historical complexity into searchable metadata.³ While this allows for general analysis it can also obscure individual experiences fueled by gaps in the archive. Historians must remain aware that databases are interpretations rather than direct proof of the past.

Reborn digital sources from the archived Web present a different set of challenges. Unlike digitized sources, websites were often born digital and archived retrospectively through services such as Internet Archive's Wayback Machine. Brügger notes that archived websites cannot be understood as exact copies of the live web as it was, but rather as reconstructed images of a bygone past. Subject to selective preservation.

² 1. Niels Brügger and Ian Milligan, eds., *The Archived Web: Doing History in the Digital Age* (Cambridge, MA: MIT Press, 2019), Page 4.

³ 3 Johnson, Jessica Marie. "Markup Bodies: Black [Life] Studies and Slavery [Death] Studies at the Digital Crossroads." *Social Text* 36, no. 4 (December 1, 2018): page 60

Historians using archived web material must study not only the content itself but the archival processes that shaped it.

Ultimately, both digitized historical databases and reborn digital web archives require that historians apply critical thinking in their methodology. However, they should not be treated identically. Digitized databases risk oversimplification of historical experience, while archived web sources raise issues concerning their reconstruction. In both cases historians must critically examine not only the content of the source itself but also the digital systems through which it has been preserved, structured and made accessible.